

HydraDB makes context stateful. HydraDG makes it auditable, perturbable, and economically measurable.

Store canonical identity once. Traverse context many times. Preserve every change, contradiction, model touch, failure, recovery, and claim boundary as graph evidence.

PAIN POINT → HYDRADB STRENGTH → HYDRADG EXTENSION

Pain point	HydraDB strength	HydraDG adds
Similarity ≠relevance	Hybrid + graph + temporal retrieval	Exact identity, custody, claim-state and evidence-path traversal.
Stale facts look like hallucinations	Versioned temporal graph	Reference → Poison → Antidote localizes first divergence and recovery.
Multi-session reasoning remains hard	Persistent cross-session context	Model/agent lineage: who introduced, inherited, corrected, or abstained.
Context volume compounds cost	Tiered storage; storage-based pricing	65.730975% deterministic identity reuse; real byte/token/call savings measured separately.
Evaluation hides inconvenient failures	Benchmarkable structured context	Positive, null, negative, timeout, blocked and abstaining states retained under claim ceilings.

DEMONSTRATED NOW

- Golden Path: 01 Reference → 02 Poison → 03 Antidote → 04 HydraDB → 05 Results → 06 Evidence → 07 Future → 08 Claim.
- Hosted HydraDB: connectivity PASS; hydradg / hydradg-judge-demo; canary PASS_REQUEST_LEVEL. Full 653 FCO / 1,692 edge parity NOT_ESTABLISHED.
- LongMemEval-S full500 K=5: 500 cases; 470 retrieval-scored; 30 abstentions; B/C/D did not beat flat A.
- Identity accounting: 31,672,976 occurrences → 10,854,020 unique → 20,818,956 duplicates → 65.730975% reuse.
- Synthetic T0/T1/T2 scoring: G*, ΔG*, Cloud Drift; later lanes N/A when no distribution is frozen.
- Context-vs-Entropy: 18,555 / 18,567 classified; 12 abstentions; 99.9354% coverage; not a Gitleaks replacement.

ECONOMICS + WHAT WE ADD

- Observed canonical Parquet footprint: 1,101,473,790 B. This is NOT a measured byte-savings figure.
- 7B duplicate-work abstraction: 2.91465384×10¹⁷ theoretical FLOPs; 0.809626 Wh theoretical equivalent; NOT measured electrical energy.
- Next KPI: Cost per Correct Governed Answer = storage + retrieval + inference + recovery cost / correct governed answers.
- Multi-agent extension: score ErrorPropagationRate, FirstDivergenceAccuracy, RecoveryRate, evidence-path recovery, context tokens and avoided model calls.
- Anticube self/non-self × safe/unsafe classification is a governance signal, not an assumed ranking boost.
- Judge takeaway: HydraDG turns HydraDB-backed memory into a falsifiable custody system — what changed, when, which agent touched it, where error entered, whether recovery worked, and what the context cost.

CLAIM BOUNDARY: identity reuse ≠byte savings · theoretical FLOPs/Wh ≠measured energy · canary readback ≠full hosted parity · SHA-256 ≠correctness/signature · BEAM/multi-agent economics remain future until receipted.